

Case Study on Information Security & I4C

Subject: Information Security

Topic: Importance of Information Security

Case Study: I4C – January 2026 Cyber Fraud Case

Mode: Individual Activity

Submission: PDF / Notebook

Time Required: 40 minutes

Submission : <https://forms.gle/ecV2uiRtdzDi0yDS8>

Learning Objective

Students will independently analyse a real-world cybercrime case and connect it with the **CIA Triad, importance of information security, security controls, and best practices.**

Case Study

In January 2026, Indian law-enforcement agencies investigated a large cyber-fraud network involving Indian SIM cards being used from Cambodia. The investigation identified thousands of suspicious mobile numbers associated with fraudulent activities, including investment and trading scams. The case demonstrated how cybercriminals can exploit mobile communication, personal information and financial systems to conduct large-scale fraud.

The **Indian Cyber Crime Coordination Centre (I4C)** plays an important role in coordinating cybercrime-related information between law-enforcement agencies and other stakeholders. I4C uses cybercrime data, analytics and coordination mechanisms to help identify suspicious activities and support investigations.

Students should study this case and analyse how **Information Security principles and practices** can help prevent or reduce such cybercrime.

Reference: [I4C – Indian Cyber Crime Coordination Centre](#)

Student Tasks

Task 1 – Understand the Case

Answer the following:

1. What happened in the case?
2. Who were the major stakeholders?
3. How were digital technologies misused?
4. What types of cyber fraud were involved?
5. Why was the case difficult to investigate?

Task 2 – Connect with Information Security

From the **10 importance points of Information Security** discussed in class, identify **at least 6 points** applicable to this case.

Complete:

Importance of Information Security	How it applies to the case
Protects Confidential Information	
Ensures Data Integrity	
Maintains Business Continuity	
Builds Customer Trust	
Prevents Financial Loss	
Supports Legal Compliance	
Protects Intellectual Property	
Prevents Cyber Attacks	
Enhances Organizational Reputation	
Enables Secure Digital Transformation	

Students may select any **6 or more**.

Task 3 – CIA Triad Analysis

Analyse the case using:

Confidentiality

- What information needed protection?
- How could criminals misuse confidential information?

Integrity

- What information or transactions could be manipulated?
- How can organisations maintain data integrity?

Availability

- Why should banking, telecom and cybercrime-reporting systems remain available?
- What could happen if these systems become unavailable?

Task 4 – Identify Security Threats

Identify **at least 5 threats** involved in or relevant to the case.

Example:

Threat	Possible Impact	Security Control
Phishing	Theft of credentials	MFA + Awareness
Social Engineering	Financial fraud	User training
Identity Misuse	Impersonation	Identity verification
Malware	Data/credential theft	Antivirus/EDR
Fraudulent Transactions	Financial loss	Transaction monitoring

Students must add **at least two additional threats** themselves.

Task 5 – Suggest Solutions

Assume you are working as a **Cybersecurity Analyst**.

Suggest **10 security measures** that could reduce such cyber fraud.

Students should include measures related to:

- MFA
- Encryption
- Strong passwords
- Access control
- Software updates
- Backup
- Security monitoring
- User awareness
- Fraud detection
- Incident response

They should explain **why each measure is useful**.

Task 6 – Think Like a Cybersecurity Analyst

Imagine that you are appointed as a cybersecurity analyst for an organisation.

Problem

Your organisation has detected:

“A large number of customers are receiving WhatsApp messages promising high returns from an investment platform. Several customers have transferred money to unknown accounts.”

Your responsibility

Prepare a **Cyber Incident Response Plan** containing:

1. Detect

How will you identify the fraud?

2. Analyse

What information will you collect?

3. Contain

What immediate actions will you take?

4. Respond

Which stakeholders should be informed?

5. Recover

How will you help affected customers and restore normal operations?

6. Prevent

What controls will you implement to prevent recurrence?

Task 7 – I4C Investigation

Visit the official I4C website and find information about **any two I4C initiatives/platforms** related to cybercrime prevention or investigation.

For each initiative, write:

1. Name of the initiative
2. Purpose
3. How it helps fight cybercrime
4. How it supports Information Security
5. One example of its practical use

[Official I4C Website](#)

Task 8 – Final Question

Write **150–200 words** answering:

“Why is Information Security important for India's digital economy?”

Your answer must include:

CIA Triad + Financial Protection + Customer Trust + Cybercrime Prevention + Legal Compliance + Digital Transformation

Submission Format

Students should prepare the assignment in the following order:

1. **Title**
2. **Case Background**
3. **Case Analysis**
4. **Importance of Information Security – Minimum 6 Points**
5. **CIA Triad Analysis**
6. **Cyber Threats**
7. **Security Solutions**
8. **Incident Response Plan**
9. **I4C Initiatives**
10. **Conclusion**
11. **References**

Submission Requirements

- Individual submission
- Recommended length: **5–8 pages**
- Use your own words.
- Use diagrams/tables wherever appropriate.
- Mention sources used for information.
- Students may use the official I4C website and other reliable sources.
- Do not simply copy information from websites or AI tools.
- Include **at least one CIA Triad diagram and one security-control diagram.**

Evaluation – 20 Marks

Criteria	Marks
Understanding of Case	3
Information Security Analysis	4
CIA Triad Analysis	3
Threat & Risk Identification	3
Proposed Security Solutions	3
I4C Research	2
Presentation & References	2
Total	20

Student Self-Check Before Submission

- I understand the January 2026 I4C case.
- I analysed at least 6 importance points.
- I explained Confidentiality, Integrity and Availability.
- I identified at least 5 threats.
- I suggested 10 security measures.
- I prepared an incident-response plan.
- I researched 2 I4C initiatives.
- I included references.
- I used my own analysis rather than copying content.